HorizonLink-SMD

argos-smd

The Argos SMD module is a small PCB (2x2cm) dedicated to act as a standalone or a coprocessor for any new design which required a satelite communication. The module is based on STM32WL55 processor and combined with GR5504 power amplifier allowing a comunication up to 27dBm. This module which pass the CLS certification is ready to be used.

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argos-smd-module

Legend

General comment such as function title, configuration, ...

Text to be added to silkscreen.

Warning text.

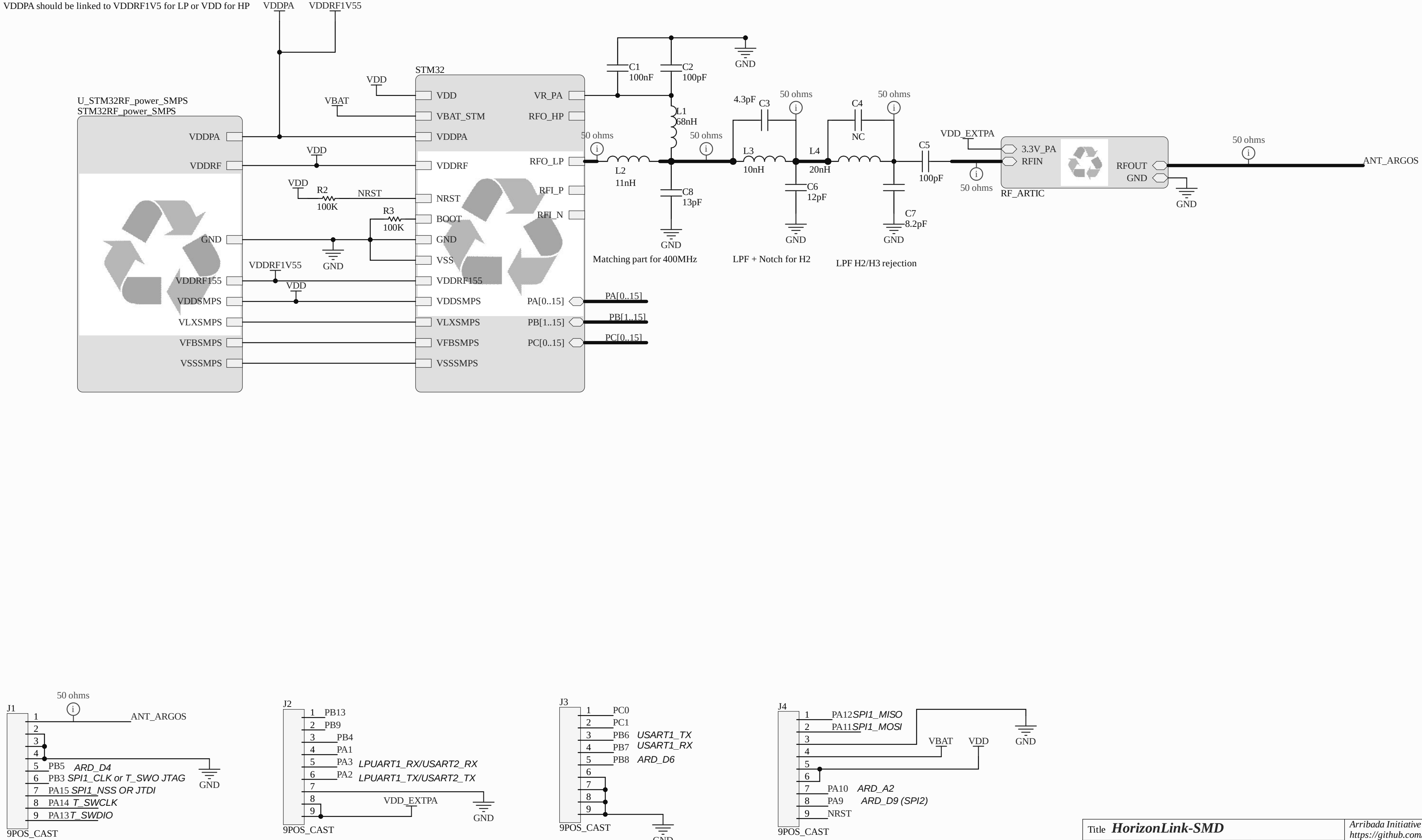
Notes to generate the board layout.

OPEN HARDWARE LICENSE AGREEMENT

Title *HorizonLink-SMD*			*Arribada Initiative* *https://github.com/arribada* *contact@arribada.org* *arribada.org* *UK - Portsmouth*
Size: A4	Reference: argos-smd	Revision: 1.0	
Date: 22/11/2024	Time: 16:59:52	Sheet 1 of 5	
File: D:\Altium\Projects\stm32wl-dev\Projects\argos-smd-hw\Project_Overview.SchDoc			

VDD = 1.8 to 3.6V : External supply for system
VDDSMPS = 1.8V to 3.6V : External supply for SMPS Step down converter
VFBSMPS = 1.45V to 1.62V (1.55v) : External supply main system regulator
VDDA = 0 to 3.6V : 1.71V DAC min, 1.8V buffer, ADC 1.62V VREFBUF = 2.24V
VDDRF = 1.8 to 3.6V :external power supply for radio (VDD)
VDDRF1V = 1.45V to 1.62V for radio
VBAT = 1.55V to 3.6V : RTC Tamp, external clock...
VREF-VREF+: input ref voltage for ADC

VDDPA should be linked to VDDRF1V5 for LP or VDD for HP



Title **HorizonLink-SMD**			Arribada Initiative https://github.com/arribada contact@arribada.org arribada.org UK - Porthmouth
Size: A3	Reference: argos-smd	Revision: 1.0	
Date: 22/11/2024	Time: 16:59:52	Sheet 2 of 5	
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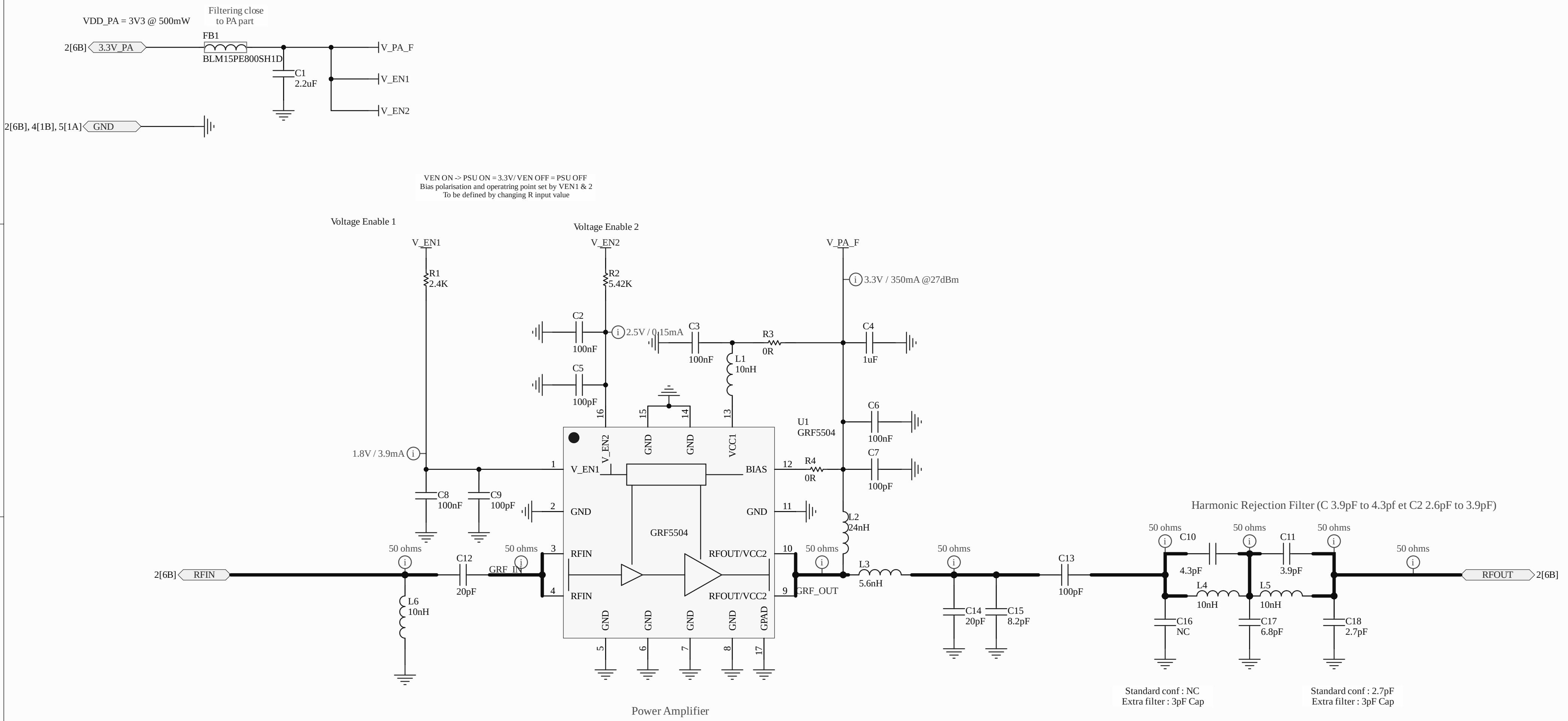
ARRIBADA initiative



Notes: Murrata
Inductor from LQG ceramic 0402
Inductor from LQW wirewound 0603
Capacitor GRM/GJM 0402

50 ohm Line

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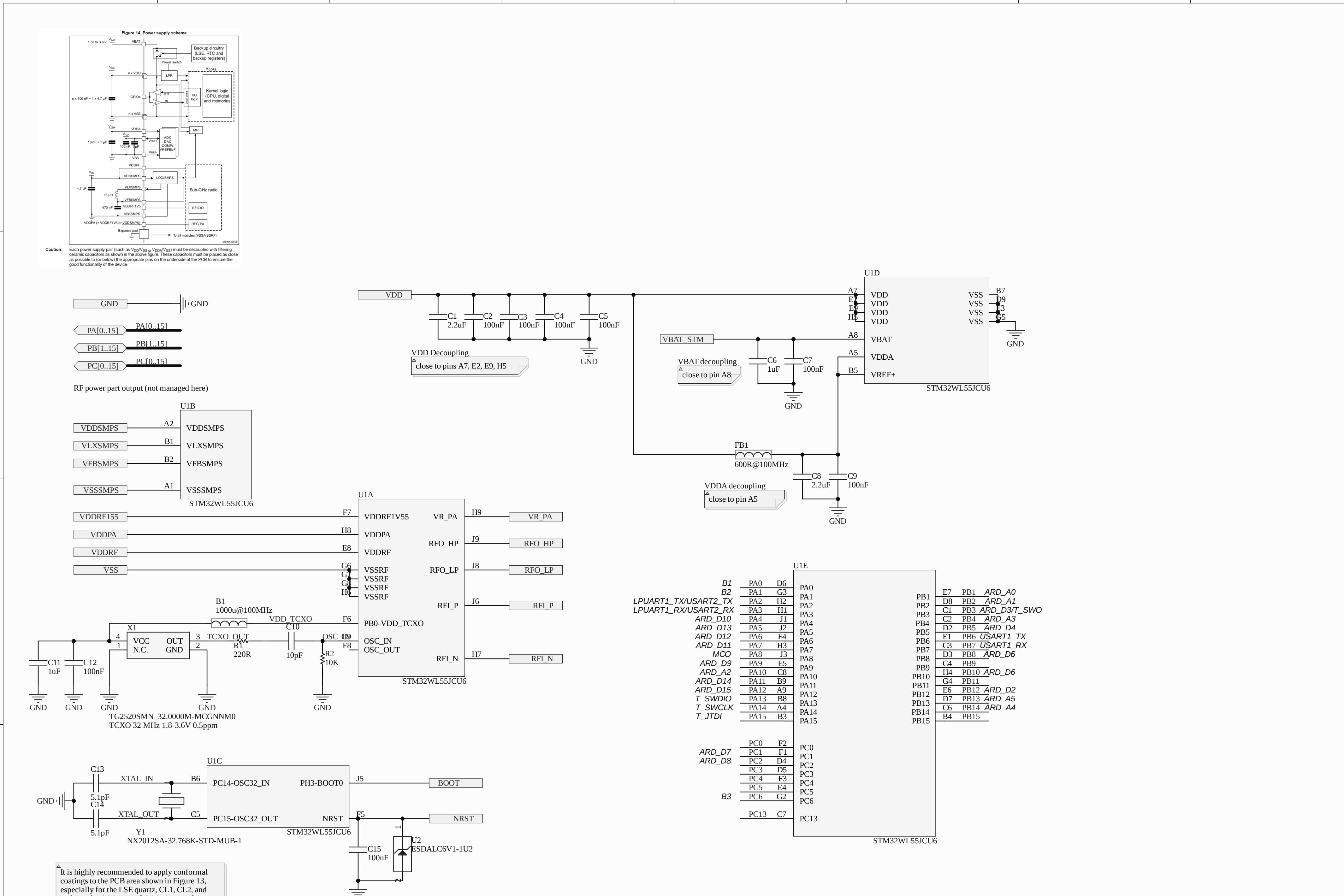

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Revision:	1.0
Date: 22/11/2024	Time: 16:59:54
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File:	D:\Altium\Projects\stm32wl-dev\hardware-lib\DeviceSheets\MCU\stm32wl55jci7-details.SchDoc

Arribada Initiative
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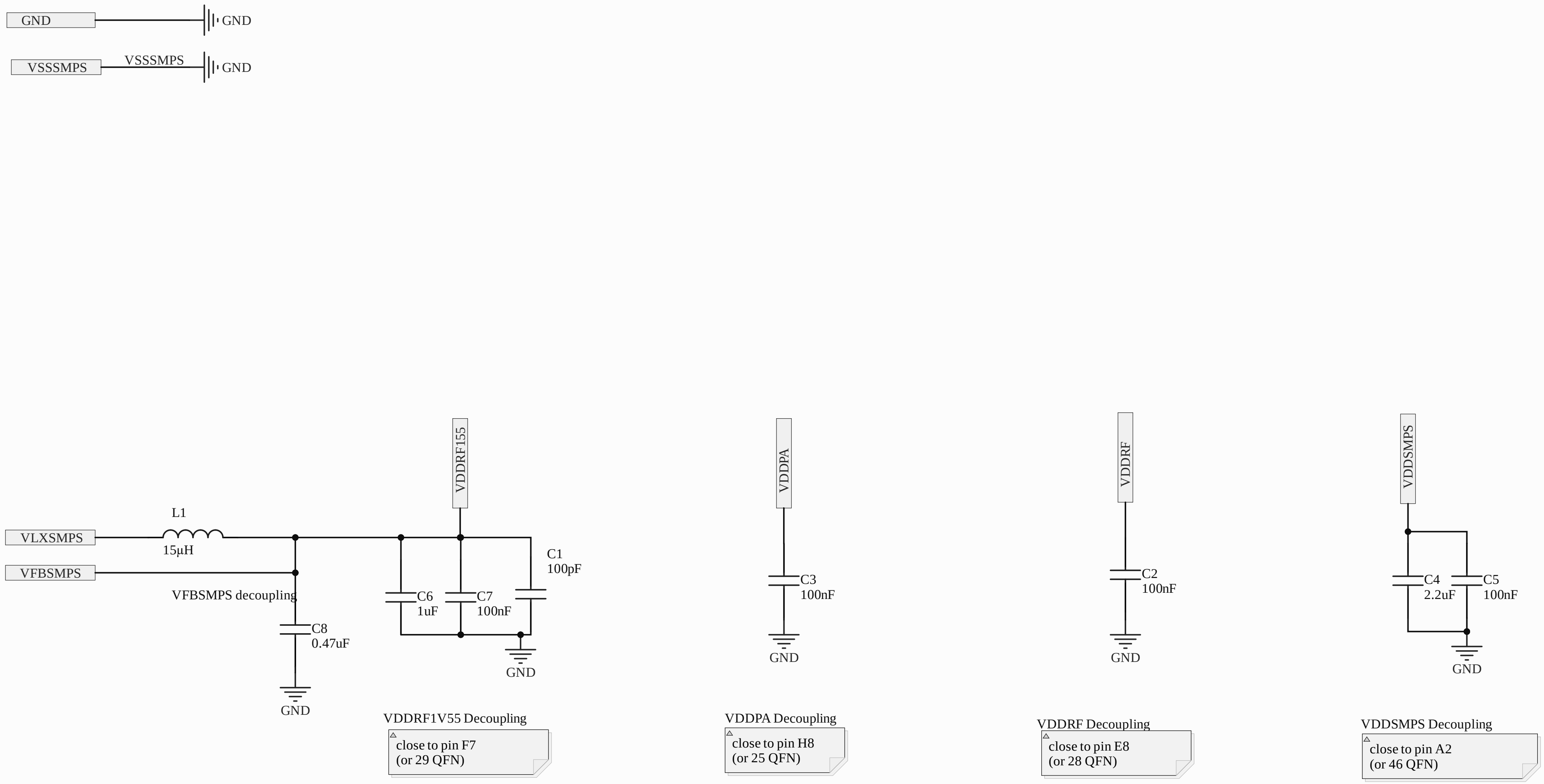

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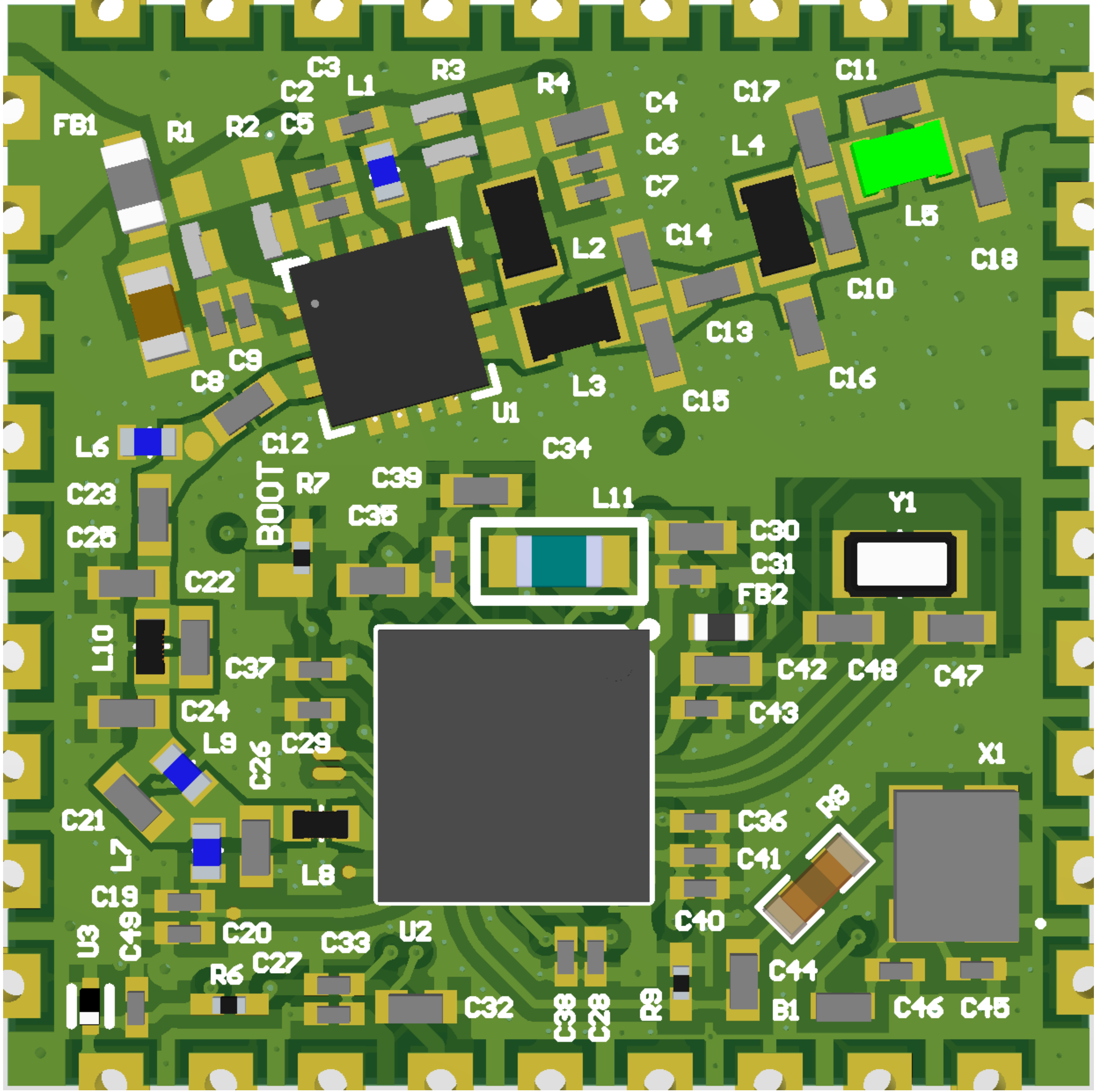
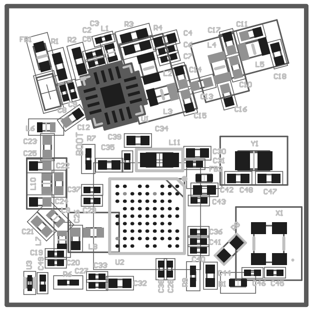
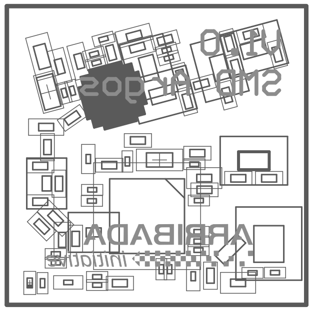

FB1
R1
R2
C5
C2
C3
L1
R3
R4
C4
C6
C7
C17
L4
C11
L5
C18
C14
L2
C10
C13
C16
C15
L3
U1
C9
C8
C12
L6
C34
C23
C25
BOOT
R7
C39
C35
L11
Y1
C30
C31
FB2
C22
L10
C37
C42
C48
C47
C24
C43
L9
C26
C29
X1
R8
C21
C36
L7
C41
L8
C19
C40
C49
C20
U3
C33
U2
C27
R6
C44
C32
C38
C28
R9
B1
C46
C45

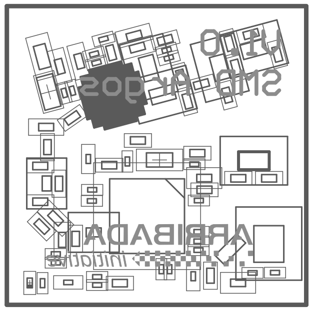


Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	Solder Resist	0,020mm	3.5	
1	Top Layer		0,035mm		
	Dielectric 1	PP-010	0,071mm	4	
2	Signal Layer 1		0,035mm		
	Dielectric 2	Core-002	1,265mm	4	
3	Signal Layer 2		0,035mm		
	Dielectric 3	PP-010	0,071mm	4	
4	Bottom Layer		0,035mm		
	Bottom Solder	Solder Resist	0,020mm	3.5	
	Bottom Overlay				

Total board thickness: 1,586mm



Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	Solder Resist	0,020mm	3.5	
1	Top Layer		0,035mm		
	Dielectric 1	PP-010	0,071mm	4	
2	Signal Layer 1		0,035mm		
	Dielectric 2	Core-002	1,265mm	4	
3	Signal Layer 2		0,035mm		
	Dielectric 3	PP-010	0,071mm	4	
4	Bottom Layer		0,035mm		
	Bottom Solder	Solder Resist	0,020mm	3.5	
	Bottom Overlay				
Total board thickness:			1,586mm		



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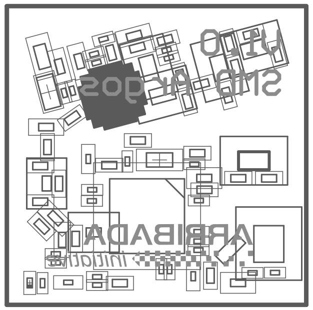
	Dielectric 3	PP-010	0,071mm	4	
4	Bottom Layer		0,035mm		
	Bottom Solder	Solder Resist	0,020mm	3.5	
	Bottom Overlay				

Total board thickness: 1,586mm



Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	Solder Resist	0,020mm	3.5	
1	Top Layer		0,035mm		
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3	Signal Layer 2		0,035mm		
	Dielectric 3	PP-010	0,071mm	4	
4	Bottom Layer		0,035mm		
	Bottom Solder	Solder Resist	0,020mm	3.5	
	Bottom Overlay				
Total board thickness:			1,586mm		

Line #	Name	Description	Designator	Quantity	Manufacturer 1	Manufacturer Part Number 1	Manufacturer Lifecycle 1	Supplier 1	Supplier Part Number 1	Supplier Unit Price 1	Supplier Subtotal 1
	1000u@100MHz	Ferrite Bead - BLM15_Z1D Series EMI Suppression Filter 1000+/-25% at 100MHz 250mA @85	B1	1	Murata	BLM15HG102SZ1D	Volume Production				
	2.2uF	Multilayer Ceramic Capacitors MLCC - SMD/SMT 2.2 uF 10 VDC 10% 0603 X7R	C1	1	Murata	GRM188R71A225KE15D	Volume Production	Mouser	81-GRM188R71A225KE15		
	100nF	Multilayer Ceramic Capacitors MLCC - SMD/SMT .1UF 16V 10% 0201	C2, C3, C6, C8, C19, C31, C33, C36, C37, C38, C39, C41, C43, C46, C49	15	Murata	GRM033Z71C104KE14D	Volume Production	Mouser	81-GRM033Z71C104KE4D		
	1uF	Multilayer Ceramic Capacitors MLCC - SMD/SMT 1.0UF 10V 10% 0402	C4, C32	2	Murata	GRM155Z71A105KE01J	Volume Production	Mouser	81-GRM155Z71A105KE1J		
	100pF	Multilayer Ceramic Capacitors MLCC - SMD/SMT 100PF 25V 5% 0201	C5, C7, C9, C20	4	Murata	GRM0335C1E101JA01D	Volume Production	Mouser	81-GRM0335C1E101JA1D		
	GRM1555C1H4R3BA01D	Multilayer Ceramic Capacitors MLCC - SMD/SMT 4.3 pF 50 VDC 0.1 pF 0402 C0G (NP0)	C10, C21	2	Murata	GRM1555C1H4R3BA01D		Digi-Key	490-6243-1-ND		
	GRM1555C1E3R9BA01D	Multilayer Ceramic Capacitors MLCC - SMD/SMT 3.9 pF 25 VDC 0.1 pF 0402 C0G (NP0)	C11	1	Murata	GRM1555C1E3R9BA01D		Mouser	81-GRM1555C1E3R9BA1D		
	20pF	Multilayer Ceramic Capacitors MLCC - SMD/SMT 20 pF 50 VDC 2% 0402 C0G (NP0)	C12, C14	2	Murata	GRM1555C1H200GA01D	Not Recommended for New Design	Mouser	81-GRM1555C1H200GA1D		
	GCM1555C1H101JA16D	Multilayer Ceramic Capacitors MLCC - SMD/SMT 100 pF 50 VDC 5% 0402 C0G (NP0) AEC-Q200	C13, C23	2	Murata	GCM1555C1H101JA16D		Mouser	81-GCM1555C1H101JA6D		
	GCM1555C1H8R2CA16J	Multilayer Ceramic Capacitors MLCC - SMD/SMT 8.2 pF 50 VDC 0.25 pF 0402 C0G (NP0) AEC-Q200	C15, C25	2	Murata	GCM1555C1H8R2CA16J	Volume Production	Mouser	81-GCM1555C1H8R2CA6J		
	CAP-NC-0402	CMS 0402 Non Connected	C16, C22	2							
	GRM1555C1E6R8BA01D	Multilayer Ceramic Capacitors MLCC - SMD/SMT 6.8 pF 25 VDC 0.1 pF 0402 C0G (NP0)	C17	1	Murata	GRM1555C1E6R8BA01D		Mouser	81-GRM1555C1E6R8BA1D		
	GRM1555C2A2R7CA01D	Multilayer Ceramic Capacitors MLCC - SMD/SMT 2.7 pF 100 VDC 0.25 pF 0402 C0G (NP0)	C18	1	Murata	GRM1555C2A2R7CA01D		Mouser	81-GRM1555C2A2R7CA1D		
	GCM1555C1H120GA16D	Multilayer Ceramic Capacitors MLCC - SMD/SMT 12 pF 50 VDC 2% 0402 C0G (NP0) AEC-Q200	C24	1	Murata	GCM1555C1H120GA16D	Volume Production	Mouser	81-GCM1555C1H120GA6D		
	GRM1555C1H130GA01D	Multilayer Ceramic Capacitors MLCC - SMD/SMT 13 pF 50 VDC 2% 0402 C0G (NP0)	C26	1	Murata	GRM1555C1H130GA01D	Not Recommended for New Design	Mouser	81-GRM1555C1H130GA1D		
	100pF	Multilayer Ceramic Capacitors MLCC - SMD/SMT 100 pF 25 VDC 1% 0201 C0G (NP0)	C27	1	Murata	GRM0335C1E101FA01J	Volume Production	Digi-Key	GRM0335C1E101FA01J-ND		
	GRM033R61C104KE14D	Multilayer Ceramic Capacitors MLCC - SMD/SMT 0201 0.1uF 16volts X5R 10%	C28, C29	2	Murata	GRM033R61C104KE14D		Mouser	81-GRM033R61C104KE4D		
	GRM155Z71A225KE44D	Multilayer Ceramic Capacitors MLCC - SMD/SMT 2.2 uF 10 VDC 10% 0402 X7R	C30, C35, C42	3	Murata	GRM155Z71A225KE44D	Volume Production	Mouser	81-GRM155Z71A225KE4D		
	0.47uF	Multilayer Ceramic Capacitors MLCC - SMD/SMT 0.47uF 10V X5R 10% SMD 0402 85°C Paper T/R	C34	1	Taiyo Yuden	LMK105ABJ474KV8F		Mouser	963-LMK105ABJ474KV8F		
	1uF	Multilayer Ceramic Capacitors MLCC - SMD/SMT 1uF+/-10% 10V X5R 06 0201	C40, C45	2	Samsung	CL03A105KP3NSNC	Volume Production				
	10pF	Multilayer Ceramic Capacitors MLCC - SMD/SMT 10 pF 50 VDC 1% 0402 C0G (NP0)	C44	1	Murata	GJM1555C1H100FB01D	Volume Production				
	5.1pF	Multilayer Ceramic Capacitors MLCC - SMD/SMT 0402 MLCC NPO 5.1 pF, +/- 0.1pF 50 V T&R GP	C47, C48	2	Walsin Technologies	0402N5R1B500CT		Mouser	791-0402N5R1B500CT		
	BLM15PE800SH1D	Ferrite Bead, 80 Ohm, 25%, 0402 Rohs Compliant: Yes	FB1	1	Murata	BLM15PE800SH1D	Volume Production	Digi-Key	490-BLM15PE800SH1DCT-ND		
	600R@100MHz	Ferrite Bead 600ohm 500mA 340mOhm 0402	FB2	1	Murata	BLM15BX601SZ1D	Volume Production				
	9POS_CAST	9 POS Castellated connector with VIA 0.6	J1, J2, J3, J4	4							
	10nH	Inductor RF Chip Multi-Layer 10nH 5% 100MHz 8Q-Factor Air 300mA 260mOhm DCR 0402 Paper T/R	L1, L6, L9	3	Murata	LQG15HS10NJ02D	Volume Production	Arrow Electronics	LQG15HS10NJ02D		
	24nH	Inductor RF Chip Unshielded Wirewound 24nH 2% 100MHz 40Q-Factor Ferrite 1.3A 0603 Paper T/R	L2	1	Murata	LQW18AN24NG80D	Volume Production	Mouser	81-LQW18AN24NG80D		
	LQW18AN5N6C80D	Inductor Wirewound , 5.6 nH, 40 Milliohm, ± 0.2nH, 6.65 GHz, 1.9 A, 0603 [1608 Metric]	L3	1	Murata	LQW18AN5N6C80D	Volume Production	Mouser	81-LQW18AN5N6C80D		
	10nH	Inductor Wirewound , 10 nH, 52 Milliohm, ± 2%, 4.75 GHz, 1.6 A, 0603 [1608 Metric]	L4, L5	2	Murata	LQW18AN10NG80D	Volume Production	Digi-Key	490-17899-1-ND		
	LQG15HN68NJ02D	Inductor RF Chip Multi-Layer 68nH 5% 100MHz 8Q-Factor Air 180mA 1.25Ohm DCR 0402 Paper T/R	L7	1	Murata	LQG15HN68NJ02D	Not Recommended for New Design	Digi-Key	490-1095-1-ND		
	LQG15WZ11NJ02D	Inductor RF Chip Multi-Layer 11nH 5% 0402 Paper T/R	L8	1	Murata	LQG15WZ11NJ02D	Volume Production	Digi-Key	490-15246-1-ND		
	LQG15WH20NJ02D	Inductor RF Chip, Aec-Q200, 20Nh, 0402, 2.1Ghz Rohs Compliant: Yes	L10	1	Murata	LQG15WH20NJ02D	Volume Production	Mouser	81-LQG15WH20NJ02D		
	MLZ1608M150WT000	Inductor RF Shielded 15µH, ±5%	L11	1	TDK	MLZ1608M150WT000	Volume Production	Mouser	810-MLZ1608M150WT000		
	2.4K	Thick Film Resistors - SMD 2.4 kOhms 62.5 mW 0402 1%	R1	1	Yageo	RC0402FR-132K4L		Mouser	603-RC0402FR-132K4L		
	5.42K	Thin Film Resistors - SMD 5.42kOhm,0402,1%,50p pm,63mW,50V	R2	1	KOA Speer	RN73H1ETTP5421F50		Mouser	660-RN73H1ETP5421F50		
	0R	Resistors Thick Film - SMD 0402 0.063 W 0 Ohm Jumper	R3, R4	2	Vishay	CRCW04020000Z0ED		Digi-Key	541-0.0JCT-ND		
	100K	Resistors Thick Film - SMD 1/20watt 100Kohms 1% 200ppm	R6, R7	2	Vishay	CRCW0201100KFNED	Volume Production				
	220R	Resistors Thick Film - SMD HP02 (0402) 1/10W 1% 220R T/R-10000	R8	1	Royal Electronic Factory	HP02WAF2200TCE		Mouser	303-HP02WAF2200TCE		
	10K	Resistor Thick Film Anti-Sulfurated Chip 0201 10kOhm 1% Paper T/R	R9	1	Yageo	AF0201FR-0710KL		Mouser	603-AF0201FR-0710KL		
	GRF5504	RF Amplifier High Efficiency, 3.5Watt PA/Saturated PA, OP1dB >34.0 dBm; 0.4 - 0.5 GHz	U1	1	Guerrilla RF	GRF5504		Mouser	459-GRF5504		
	STM32WL55JCI7_details	Multiprotocol Wireless 32-bit MCU with LPWAN Sub-GHz radio solution(BGA)	U2	1	STMicroelectronics	STM32WL55JCI7		Mouser	511-STM32WL55JCI7		
	ESD_6V	ESD protection 6V1 - Varistors MLV 0603 18V 3pF	U3	1	STMicroelectronics	ESDALC6V1-1U2	Volume Production	Mouser	511-ESDALC6V1-1U2		
	TG2520SMN_32.0000M-MCGNNM0	TCXO Oscillators TG2520SMN 32.0000M-MCGNNM0: TCXO 2.8V 3.3V 0.5PPM -40 85C NON O/E 1K TR	X1	1	Epson	TG2520SMN32.0000M-MCGNNM0		Mouser	732-2520SMN32MCGNNM0		
	NX2012SA-32.768K-STD-MUB-1	Crystals 32.768KHZ 12.5PF SMD	Y1	1	NDK	NX2012SA-32.768K-STD-MUB-1	Volume Production	Mouser	344-NX2012SA32768K1		